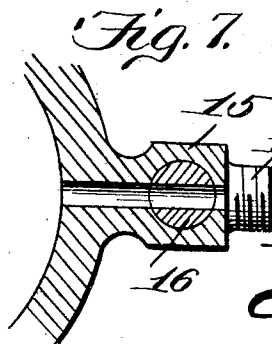
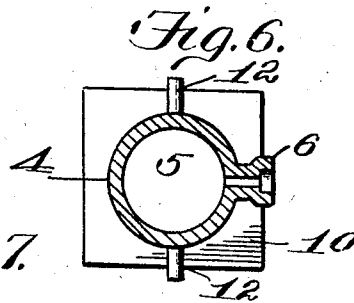
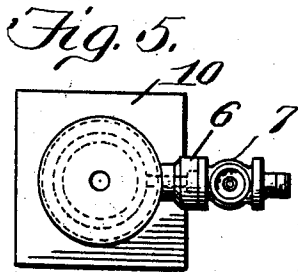
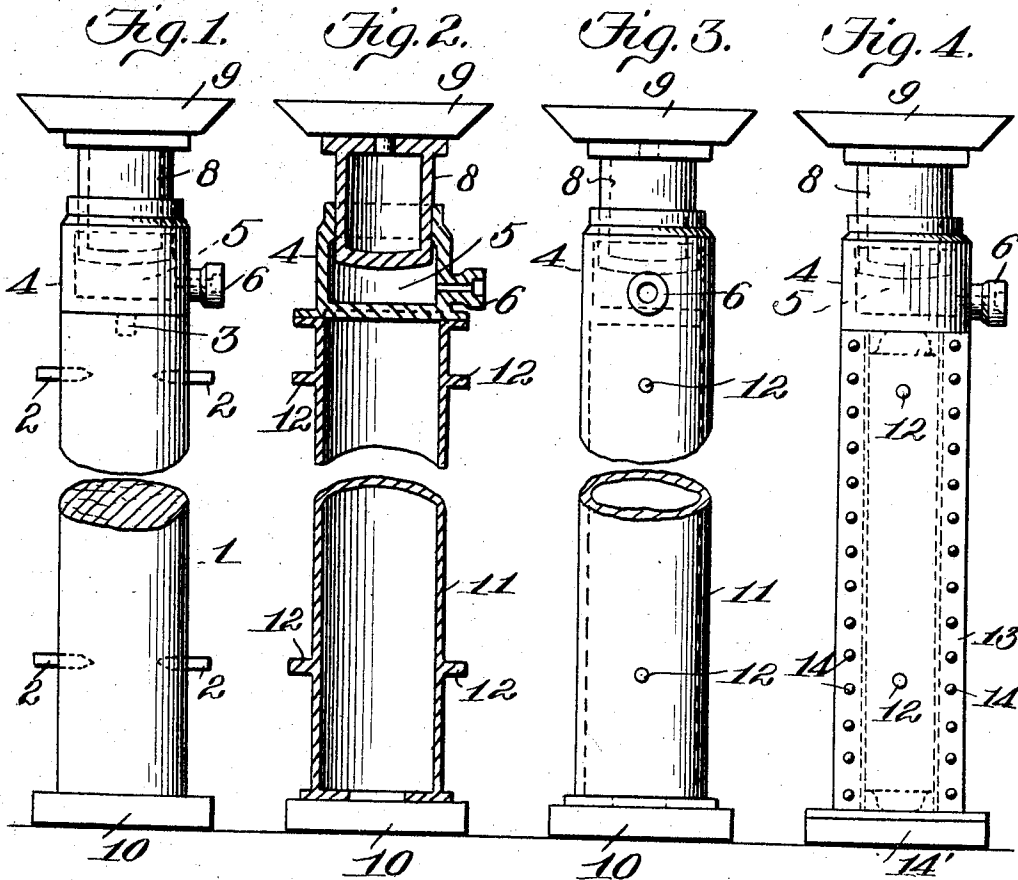


No. 771,057.

PATENTED SEPT. 27, 1904.

H. S. GAY.
PORTABLE POST.
APPLICATION FILED MAR. 18, 1904.

NO MODEL.



Witnesses:
C. D. Kessler,
James L. Norring.

Inventor
Harry S. Gay
By
James L. Norring
Atty.

UNITED STATES PATENT OFFICE.

HARRY S. GAY, OF SHAMOKIN, PENNSYLVANIA.

PORTABLE POST.

SPECIFICATION forming part of Letters Patent No. 771,057, dated September 27, 1904.

Application filed March 18, 1904. Serial No. 198,741. (No model.)

To all whom it may concern:

Be it known that I, HARRY S. GAY, a citizen of the United States, residing at Shamokin, in the county of Northumberland and State of Pennsylvania, have invented new and useful Improvements in Portable Posts, of which the following is a specification.

This invention relates to certain new and useful improvements in portable posts particularly adapted for supporting the overhanging walls or roofs when mining, although the post is adapted for any use for which it is applicable.

The invention further aims to provide a portable post provided with means which will permit of adjusting the height of the post and also to permit of removing the post when occasion requires.

In the method of extracting ores and minerals from their native beds it is frequently necessary, for the purpose of maintaining means of egress and ingress for the continuation of the work and for the protection of life, to support the overhanging walls or roofs by means of posts or props. These posts or props are not adjustable, and in order to fit them closely and rigidly in position it is necessary to remove a part of the roof or floor or more frequently to fill the intervening space between the top of the post or prop and the roof with wedges, caps, or other devices by firmly driving them in place. Consequently there is no provision made to facilitate the removal of the posts or props when it is desirable or necessary to do so. In fact, the operation of removing the posts or props is laborious and costly, and often it is dangerous and at times it is impossible to move the posts or props owing to the sinking of the overhanging walls or roofs. Therefore this invention aims to provide a new and novel form of post which will permit of its removal and also permit of its being adjusted to rigidly retain it in position when occasion requires.

The invention further aims to provide a portable post for mining and other usages which shall be extremely simple in its construction, strong, durable, efficient in its use,

and comparatively inexpensive to manufacture and set up.

With the foregoing and other objects in view the invention consists of the novel combination and arrangement of parts hereinafter more specifically described, illustrated in the accompanying drawings, and particularly pointed out in the claims hereunto appended.

In describing the invention in detail reference is had to the accompanying drawings, forming a part of this specification, wherein like reference characters denote corresponding parts throughout the several views, and in which—

Figure 1 is a side elevation of a portable and adjustable post constructed in accordance with my invention. Figs. 2 and 3 are respectively a sectional elevation and a side elevation of a modified form of post constructed in accordance with my invention. Fig. 4 is a side elevation of a modified form of post constructed in accordance with my invention. Fig. 5 is a top plan view of any form of post, showing the coupling for establishing communication between the chamber in the upper end of the post and a fluid-supply. Fig. 6 is a sectional plan of any form of post, showing the fluid-chamber; and Fig. 7 is a sectional detail of a modified form of inlet-opening for the fluid-chamber.

Referring to Fig. 1 of the drawings, 1 denotes the body of the post, which is solid and constructed, preferably, of wood and is provided with a plurality of grips 2 to permit of handling the post. One end of the body portion 1 has secured thereto in any suitable manner, as at 3, a casing 4, which forms a fluid-chamber 5. The casing 4 is provided with a screw-threaded inlet-nipple 6. To the latter is connected a valve-coupling 7, which communicates with a source of fluid-supply. Preferably the valve-coupling 7 is connected with the fluid-supply through the medium of flexible hose. The casing 4 is open at its top, and within the chamber 5 operates a plunger 8, against which is placed a cushion 9. As shown, the lower end of the plunger is slightly rounded. The cushion 9 may be of any suitable material. The body portion 1 is preferably

mounted upon the end block 10, which is of any suitable material.

In the form of post shown in Figs. 2, 3, and 4 the adjustable parts thereof are constructed in the same manner as that shown by Fig. 1, and consequently the same reference characters are applied thereto. In the form of the post shown in Figs. 2 and 3 an end block of the same character as that shown in connection with Fig. 1 is also employed. The differences between the structure shown in Figs. 2 and 3 and that of Fig. 1 is that in Figs. 2 and 3 the body portion is hollow and constructed of a vertically-extending metallic cylinder 11, provided with integral grips 12. As to Fig. 4, the body portion is constructed of a plurality of separate metallic sections 13, suitably bolted or riveted together, as at 14, and which forms the body portion of the post hollow. The ends of the sections 13 are enlarged, as at 14', to form an integral end block for the same purpose as the end blocks 10.

In Fig. 7 is shown a modified construction of inlet-nipple, and in this connection the nipple 15 is provided with the cut-off 16 and may be screw-threaded exteriorly or interiorly, as shown exteriorly, as at 17, to receive the end of the fluid-supply pipe.

When any of the forms of the post are adapted to be used, they are placed under the roof or overhanging wall which is to be supported, the end block placed at the bottom of the body portion of the post, and the cushioning device mounted upon the top of the plunger 8. The fluid-supply pipe is connected to the inlet-nipple and the valve is opened, so that the fluid can elevate the plunger to the proper height until the post is rigidly secured in position. It may be now stated that the fluid employed is preferably water. The supply of fluid is then cut off, and the fluid-supply pipe is then disconnected and the operation repeated for any number of similar posts. When it is desired to remove the post, the valve is opened, which will permit of the fluid exhausting from the chamber 5 and allowing the plunger to lower. Consequently the post can be removed without any trouble to the position where it is to be employed for again supporting the roof or overhanging wall. It will also be evident that the post can be used in a reverse manner to that as stated. For example, the cushion 9 may be placed on the floor and the post mounted thereon, with the plunger 8 resting on the cushion 9 and the end block 10 interposed between the post and the overhanging wall or roof. When the post is used in the manner as just set forth, the end block 10 or 14' becomes the head-block and the cushion 9 the base-block, but when the post is used so that the plunger will be arranged at the upper end thereof the block 10 becomes the base-block and the cushion 9 the head-block.

It is thought that the many advantages of

the portable and adjustable post used for mining purposes can be readily understood from the foregoing description, taken in connection with the accompanying drawings, and it will furthermore be evident that changes, variations, and modifications can be resorted to without departing from the spirit of my invention or sacrificing any of its advantages, and I therefore do not wish to restrict myself to the details of construction hereinbefore described, and set forth in the annexed drawings, but reserve the right to make such changes, variations, and modifications as come properly within the scope of the protection prayed.

Having thus fully described my invention, what I claim as new, and desire to secure by Letters Patent, is—

1. A portable and adjustable post comprising a body portion, a casing permanently connected with the top thereof and forming a fluid-chamber, a vertically-adjustable plunger operating in said chamber, and means for connecting said chamber with a fluid-supply to vertically adjust the plunger.

2. A portable and adjustable post comprising a body portion, a casing suitably connected with the top thereof and forming a fluid-chamber, a vertically-adjustable plunger operating in said chamber, means for connecting said chamber with a fluid-supply to vertically adjust the plunger, a base for the bottom of said body portion, and a cushioning means mounted upon the piston.

3. A portable and adjustable post comprising a body portion, a vertically-adjustable means carried thereby for increasing the height of the post, and a plurality of grips suitably connected with the body portion of the post.

4. A portable and adjustable post comprising a hollow body portion, a vertically-adjustable plunger carried by the upper end thereof and adapted to increase the height of the post, and means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger.

5. A portable and adjustable post comprising a body portion, a vertically-adjustable plunger permanently carried by the upper end thereof and adapted to increase the height of the post, and means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger.

6. A portable and adjustable post comprising a hollow body portion, a vertically-adjustable plunger carried by the upper end thereof and adapted to increase the height of the post, means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger, and a cushioning means mounted upon the upper end of said plunger.

7. A portable and adjustable post comprising a body portion, a vertically-adjustable plunger carried by the upper end thereof and

adapted to increase the height of the post, means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger, and a cushioning means mounted upon said plunger.

8. A portable and adjustable post comprising a hollow body portion, a vertically-adjustable plunger carried by the upper end thereof and adapted to increase the height of the post, means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger, and grips suitably connected with the body portion.

9. A portable and adjustable post comprising a body portion, a vertically-adjustable plunger carried by the upper end thereof and adapted to increase the height of the post, means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger, and grips suitably connected with the body portion.

10. A portable and adjustable post comprising a body portion, a vertically-adjustable plunger carried by the upper end thereof and adapted to increase the height of the post, means for supplying and exhausting a fluid to and from the upper end of the post to adjust the plunger, a cushioning means mounted upon said plunger, and a base for the body portion.

11. A portable and adjustable post comprising a body portion, and a fluid-operated means secured to one end of the body portion and adapted to increase the length of the post.

12. A portable and adjustable post comprising a hollow body portion, and a fluid-operated means secured to the body portion and adapted to increase the length of the post.

13. A portable and adjustable post comprising a body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, and grips suitably connected with the body portion.

14. A portable and adjustable post comprising a hollow body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, and grips suitably connected with the body portion.

15. A portable and adjustable post comprising a body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, and a cushioning means mounted upon said fluid-operated means.

16. A portable and adjustable post comprising a body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, a cushioning means mounted upon said fluid-operated means, and a base for said body portion.

17. A portable and adjustable post comprising a hollow body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, a cushioning means carried by said fluid-operated means, and a base for said body portion.

18. A portable and adjustable post comprising a body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, a cushioning means mounted upon said fluid-operated means, a base for said body portion, and grips suitably connected with the body portion.

19. A portable and adjustable post comprising a hollow body portion, a fluid-operated means secured to the body portion and adapted to increase the length of the post, a cushioning means carried by said fluid-operated means, a base for said body portion, and grips suitably connected with the body portion.

20. A portable and adjustable supporting-post comprising a body portion, a casing permanently connected with one end of the body portion and forming a fluid-chamber, a vertically-adjustable plunger operating in said chamber, said body portion, casing and plunger bodily movable together, and detachable means for connecting said chamber with a fluid-supply to vertically adjust the plunger.

In testimony whereof I have hereunto set my hand in presence of two subscribing witnesses.

HARRY S. GAY.

Witnesses:

NAAMAN JACKSON,
J. E. BURLEND.